# IGCSE 0607 Practice Questions — E3.2

## Sketching graphs on a calculator

**Total Questions:** 9 — **Total Marks:** 73 Marks

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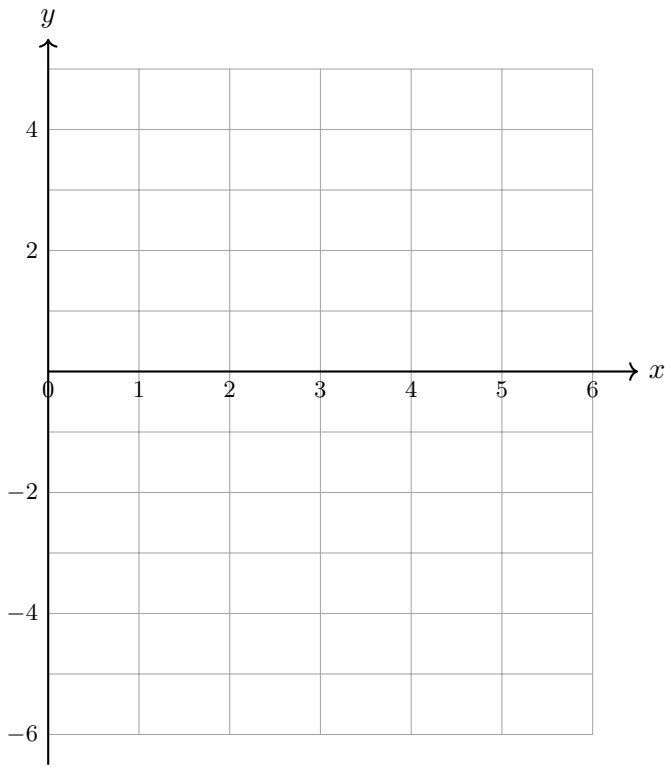
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- **E3.2** Sketching graphs on a calculator
  → **Paper 4 (Calculator)** *(9 Qs)*


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**1** Let $f(x) = -x^2 + 6x - 5$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$ for $0 \leq x \leq 6$.




**[2]**

**(b)** Find the zeros of $f(x)$.

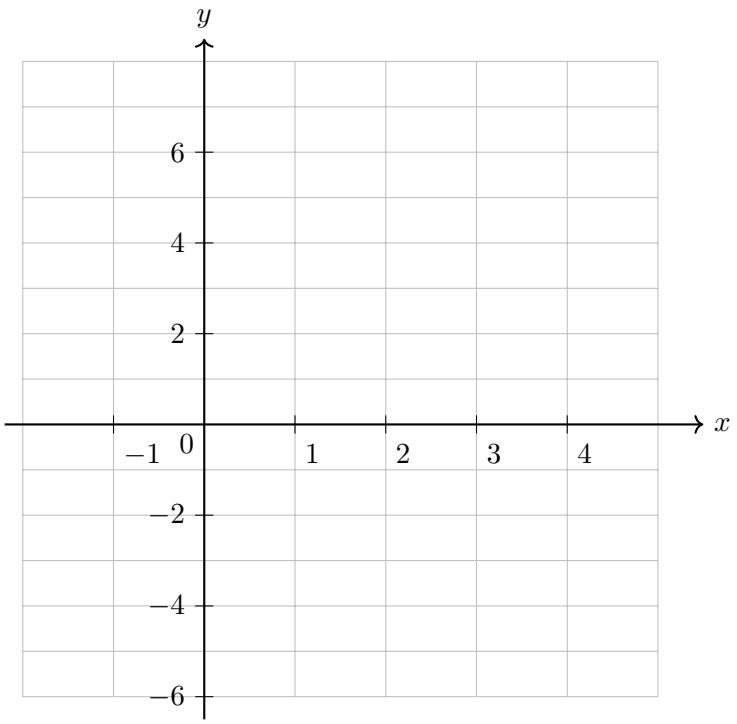
$x =$ ..... , $x =$ ..... **[2]**

**(c)** Find the coordinates of the vertex of the graph.

( ..... , ..... ) **[2]**

**2** Let $f(x) = -2x^2 + 5x + 3$.

**(a)** On the axes, sketch the graph of $y = f(x)$ for $-1.5 \leq x \leq 4$.




**[2]**

**(b)** Find the coordinates of the local maximum.

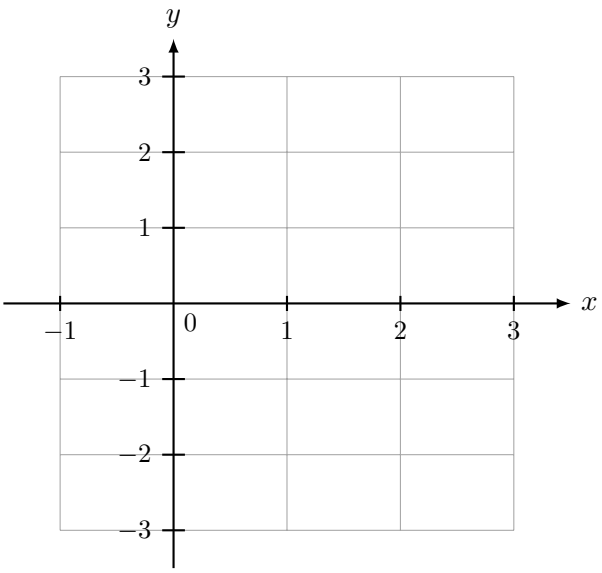
( ..... , ..... ) **[2]**

**(c)** Find the zeros of $f(x)$.

$x =$ ..... , $x =$ ..... **[2]**

**3** Consider the function $f(x) = x^3 - 3x^2 + 2$ for $-1 \leq x \leq 3$.

**(a)** On the grid, sketch the graph of $y = f(x)$.




**[2]**

**(b)** Find the zeros of $f(x)$.

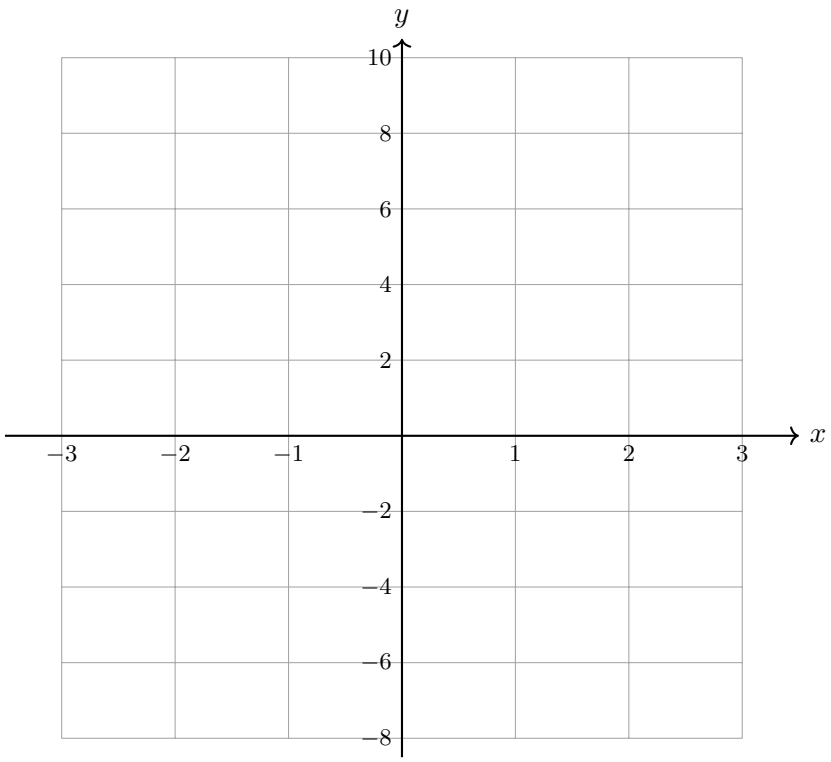
$x =$ ..... **[2]**

**(c)** Find the coordinates of the local minimum point of the graph.

( ..... , ..... ) **[2]**

**4** Let $f(x) = x^3 - 3x + 1$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.




**[2]**

**(b)** Find the coordinates of the local maximum point.

( ..... , ..... ) **[2]**

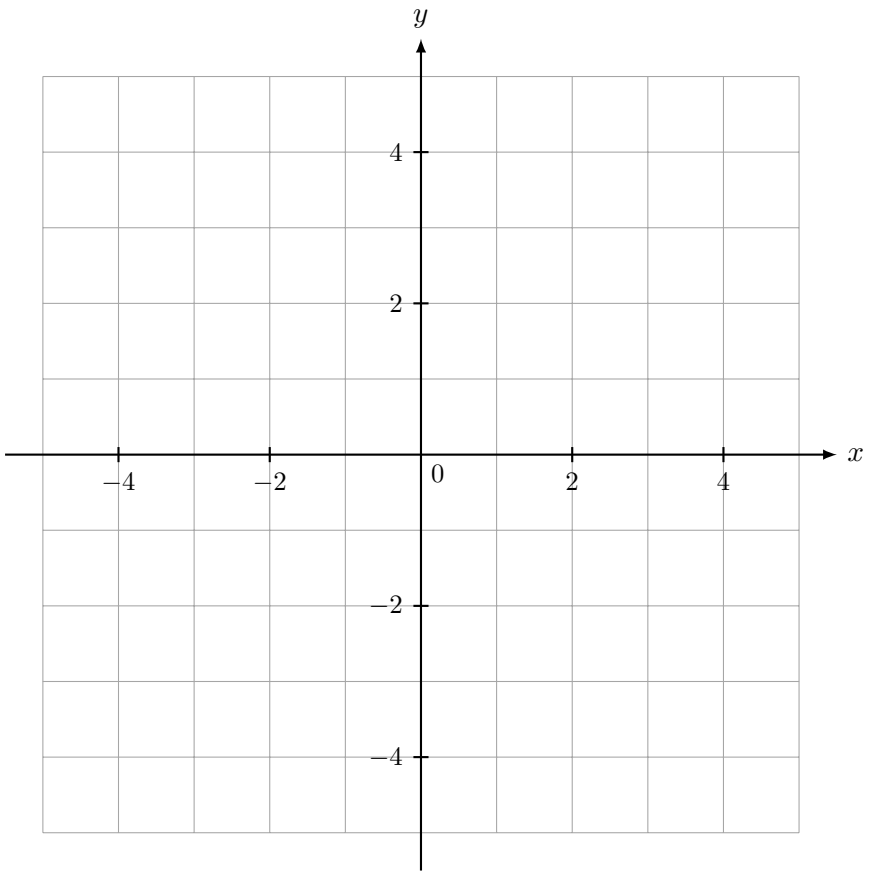
**(c)** On the same axes, sketch the graph of $y = 2x - 1$. **[1]**

**(d)** Find the $x$-coordinates of the points of intersection of the two graphs.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[2]**

**5** Let $f(x) = \dfrac{2}{x+1}$ and $g(x) = x - 2$ for $-5 \leq x \leq 5$.

**(a)** On the grid, sketch the graphs of $y = f(x)$ and $y = g(x)$.




**[3]**

**(b)** Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

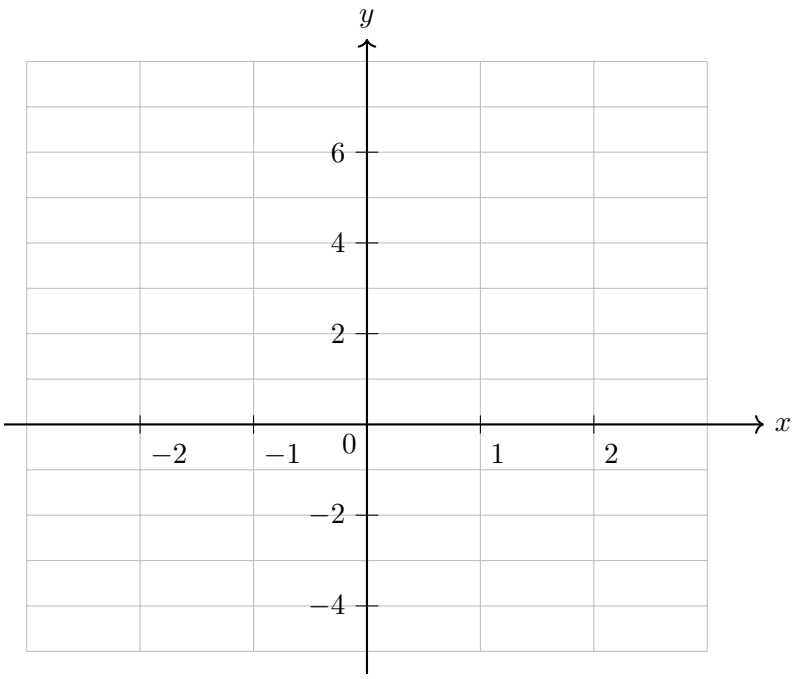
..... **[1]**

**(c)** Find the coordinates of the points of intersection of the two graphs.

Point 1 = ..... , Point 2 = ..... **[3]**

**6** Let $f(x) = x^3 - 3x + 2$.

**(a)** On the axes, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.




**[3]**

**(b)** Find the coordinates of the local maximum.

( ..... , ..... ) **[2]**

**(c)** Find the zeros of $f(x)$.

$x =$ ..... , $x =$ ..... **[2]**

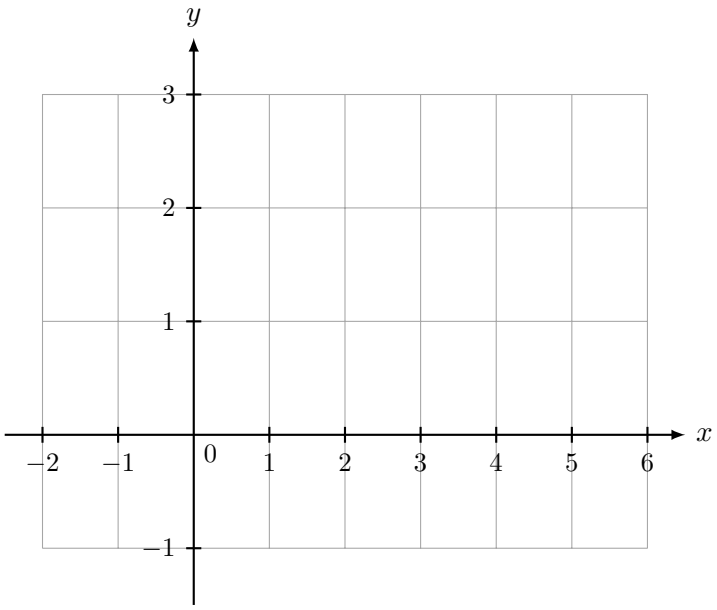
**(d)** On the same diagram, sketch the graph of $y = 2x + 1$. **[2]**

**(e)** Find the $x$-coordinates of the points of intersection of $y = f(x)$ and $y = 2x + 1$.

..... **[3]**

**7** The function $f(x)$ is defined as $f(x) = \dfrac{x^2}{2^x}$ for the domain $-1 \leq x \leq 5$.

**(a)** On the grid, sketch the graph of $y = f(x)$.




**[3]**

**(b)** Find the coordinates of the local maximum point.

( ..... , ..... ) **[2]**

**(c)** Find the $x$-coordinates of the points where $f(x) = 0.5$.

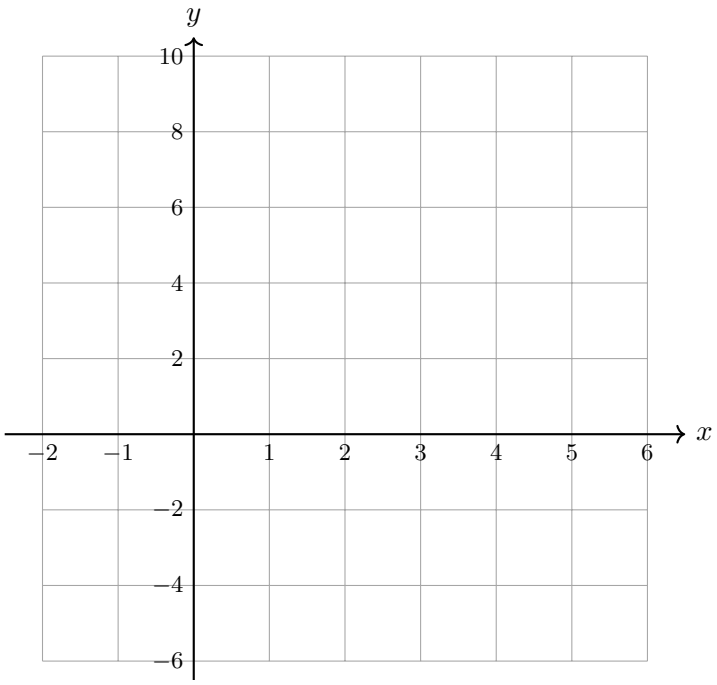
$x =$ ..... **[2]**

**(d)** Write down the range of values of $k$ for which the equation $f(x) = k$ has exactly 3 different solutions.

..... **[2]**

**8** Let $f(x) = \dfrac{x^2 - 2x + 4}{x - 2}$.

**(a)** On the axes provided, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 6, x \neq 2$.




**[3]**

**(b)** Write down the equation of the vertical asymptote.

..... **[1]**

**(c)** Find the coordinates of the local maximum point.

( ..... , ..... ) **[2]**

**(d)** Find the coordinates of the local minimum point.

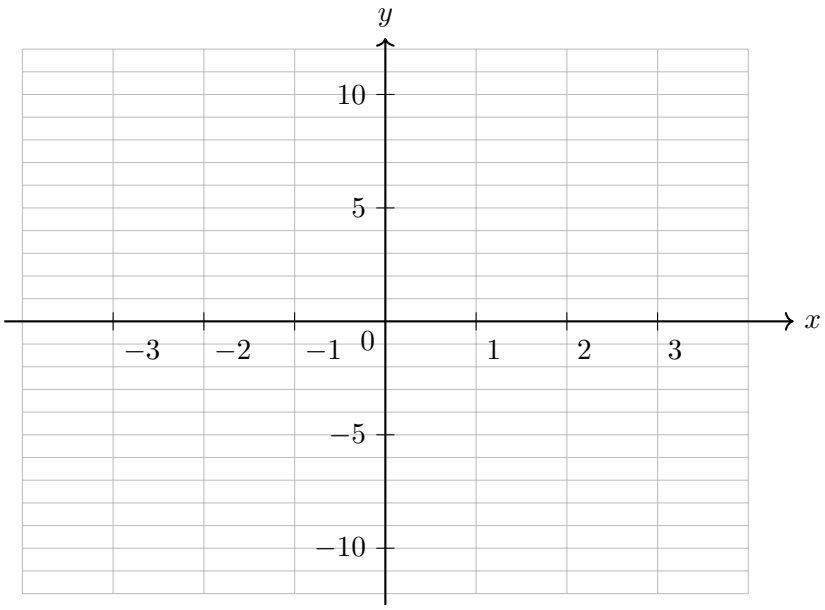
( ..... , ..... ) **[2]**

**(e)** The equation $f(x) = k$ has no real solutions.
Write down the range of values of $k$.

..... **[2]**

**9** Let $f(x) = \dfrac{x^3 + 2}{x}$.

**(a)** On the axes, sketch the graph of $y = f(x)$ for $-3 \leq x \leq 3$.




**[3]**

**(b)** Find the coordinates of the local minimum.

( ..... , ..... ) **[2]**

**(c)** Find the zero of $f(x)$.

$x =$ ..... **[1]**

**(d)** Write down the equation of the vertical asymptote.

..... **[1]**

**(e)** Solve the equation $f(x) = 4$.

..... **[3]**